EXPRESSIVE PARAMETERS FOR MUSICAL ROBOTS

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ABSTRACT

Expressive musical robots are those that, through carefully-chosen parameters, afford composers a relatively wide range of compositional options. This paper introduces a number of robots designed with parametric expressivity in mind: robotic chordophones, percussion instruments, and keyboard instruments are presented, and the mechatronic means by which their expressive parameters are realised are discussed.

Such parametrically-dense instruments can pose a challenge to use: a musical robot-specific suite of networking software is presented, intended to allow for easier-to-use complicated musical robots. In addition to the presentation of new musical robots, a history of preceding and contemporary expressive musical robotic systems is provided.

1. INTRODUCTION:

Performance-oriented musical robots are instruments, conceived of as tools for extending the compositional terrain able to be explored by composers, artists, and performing musicians. As musical instruments, they must be capable of interfacing between the artist's musical wishes and the physical world. One way to do this is to present the artist with a vast array of limited-capability devices. A second approach is to create a smaller number of robotic instruments, each able to convey a relatively wide range of musical ideas. The second approach is what is focused upon in this paper: the design, building, and application of expressive musical robots.

Why is expressivity a goal worth striving for in the design of musical robotics? Robotic instruments bring computer music into physical space. Their physicality is what makes them compelling, and the reasons for this are twofold: sonically, the robot shares the space with the listener; visually, the contours and shifts present within the structure of electronically-composed music can be visually coupled with an object. By creating objects with many parameters, each capable of predictable and precise manipulation of the instrument's audio output, those two reasons are accentuated: the visual and aural appeal of musical robots increases along with their mechatronic expressivity.

Expressivity is a word that appears many times in this paper. As a potentially subjective word, a definition of expressivity is in order. In this context, expressivity refers to the ability of a mechatronic musical system to affect a wide range of musical parameters. In essence, according to synthetic instrument builders Xavier Rodet and Christophe Vergez, an instrument capable of "subtle nuances or accentuated features which are under the control of the musician allows for musical expressivity and phrasing" (Rodet, 1999).

As little has been written specifically concerning expressive robotics, a roboticist may turn to other music writing to learn more about the subject. Patrik N. Juslin et al., in "Feedback Learning of Musical Expressivity" (Juslin, 2004), engage in a demystification of musical expressivity in all aspects of music, from compositionally-based to instrument-derived sources. Their work serves to provide useful criteria for any instrument builder concerned with providing musicians and composers with affordances for expressive use. Juslin and his co-authors subdivide musical expression into five categories: piece-related, performer-related, listener-related, context-related, and instrument-related. Juslin states that the factors affecting instrument-related expressivity include "acoustic parameters available, Instrument-specific aspects of timbre, pitch, etc., [and] technical difficulties of the instrument" (Juslin, 2004). The work presented in this paper aims to work from those three parameters (as well as those described by Cornelius Poepel in (Poepel, 2005)), focusing on the creation of musical robots endowed with arrays of acoustic and timbral parameters while remaining usable by human composers, musicians, and artists.

After an overview of existent expressive musical robotic chordophones, keyboard instruments, and percussion systems, four recently completed instruments are presented. The instruments' mechatronic subsystems are focused upon: details are provided concerning the means by which expressive sounds are created through mechanical and electronic systems. These instruments are described, with a focus on their expressive parameters. Finally, the ways in which human composers may interface with the new instruments are

discussed, and opportunities for subsequent work are presented.

2. RELATED WORK

While many of the challenges of early mechanical musical instruments (such as those described in (Ord-Hume, 1983)) were likely centred upon a need to allow for the repeatable playback of musical events (with parametrically-rich expressivity taking a back seat to this goal), the latter half of the 19th Century saw a maturation of the technology. With maturing mechanical music systems, inventors began to add additional parameters to their instruments: for example, (*Science*, 1888) describes fully mechanical chordophones capable of tremolo effects.



Figure 1. Trimpin with sound sculpture *Conloninpurple*

The advent of the loudspeaker largely precipitated the end of rapid innovations in mechanical music systems, and it was not until the 1970's that artist/engineers began to re-explore mechanical music, integrating electronic components into their new instruments. The decline of automatic music and the subsequent ascendency of mechatronic music is described in (Murphy, 2012), which introduces sound artists Trimpin (Leitman, 2011) (see Figure 1) and Godfried-Willem Raes as early innovators in expressive musical mechatronics (Maes, 2011). The following subsections detail other recent innovations in expressive musical robotic systems.

2.1. Recent Advances in Robotic Chordophones

Robotic guitars, bass guitars, violins, and other chordophones have ancestors in pre-loudspeaker automatic instruments such as the Banjorchestra and the varieties of Orchestrions popular in the 19th century. These instruments typically augmented existing human-playable instruments by placing them beneath pneumatic plucking mechanisms and string fretting systems. As with other automatic instruments, these devices largely fell out of favour with the advent of the phonograph and, subsequently, the loudspeaker.

Modern innovations in robotic chordophones fall into a two main categories: expressive pitch-shifting techniques and expressive string actuation techniques.

There are two main methods of pitch shifting (also called fretting) on musical robots: arrays of solenoids and linear motion-based approaches. Fretters consisting of arrays of solenoids (such as those employed by Trimpin on his 2007 *Jackbox* or Godfried-Willem Raes on his 2011 Synchochord instrument) function by pinching the string against the guitar's neck; such approaches require a fretboard and are of relatively low resolution and latency. To allow for higher resolution (and more expressive control over an instrument's pitch output), a number of artist/engineers have developed linear motion-based fretting systems. Nicholas Anatol Baginsky's Aglaopheme robot (1992) is an early example of such a system, featuring a linear-motion slide capable of clamping against the string at various points along its travel, changing the string's pitch with a much higher resolution than can discrete solenoid arrays. (Murphy, 2013) describes Baginsky's Aglaopheme in greater detail. A second similar system is Eric Singer's GuitarBot (Singer, 2003), a modular guitar robot capable of even higher pitch expressivity than the Aglaopheme: as shown in Figure 2, each string has a separate linear motion slide, allowing for individual control over each string's pitch. The new works presented below in this paper build upon the continuous linear positioning systems of Baginsky and Singer.



Figure 2. Eric Singer's GuitarBot

Just as two main methods are used to fret the strings, there are two main techniques for picking the strings of robotic chordophones: reciprocating solenoid pickers and motorised pickwheels. Trimpin's JackBox uses both: reciprocating solenoids alternately drag a pick back and forth across a string; a pickwheel consists of a rotary motor (attached to a DC motor or stepper

motor) with picks axially mounted to its shaft. As the pickwheel spins, the picks are dragged across the string, actuating it. While neither method allows for a high degree of precise expressive control over the string actuation events, the rotary pickwheel system is built upon in this work because of its inherent high speed and ease of implementation.

(Huang, 2012) and (Ferguson, 2013) present other methods of string actuation: the bowing mechanism shown in (Huang, 2012) is mechatronically complicated and, while potentially quite expressive, outside the scope of this research. The feedback-based system described in (Ferguson, 2013) could be useful to integrate into existing systems, allowing for an additional method of string actuation.

While many existent mechatronic chordophones are variations on those presented here, the works described above are notable in that they strove to increase the expressivity of robotic guitars and basses. The new works presented below are inspired by these existent works, largely focusing on the application of more actuators to the mechatronic subsystems, allowing composers more degrees of expressive freedom.

2.2. Recent Advances in Robotic Keyboard Players

Two parallel trends are evident in studying recent automatically-actuated keyboard instruments: that of assistive augmented pianos and that of fully-automated instruments. Assistive augmented pianos consist of instruments created to be played by a human performer; the performer's real-time playing is then altered through mechatronic means. Conversely, fully-automated instruments are capable of audio playback with no direct human interaction. Occupying a middle ground between assistive and automated keyboard instruments are the Yamaha Disklavier and similar instruments, playable both interactively and autonomously.

While not new (Ord-Hume, 1985), recent mechatronic keyboard playing aids are capable of high degrees of expressivity. Edgar Berdahl et al. in (Berdahl, 2005) present a piano augmented with electromechanical actuators. Subsequently, Michael Fabio created a similar purpose-built instrument designed to be human-played (Fabio, 2007). Most recently, Andrew McPherson's Resonator Piano adds the ability to perform pitch-bend, tremolo, and other untraditional playing techniques on augmented pianos (McPherson, 2010).

A common theme of the above instruments is that they are designed to be played in real time by a human performer. A second avenue of mechatronic instrument development focuses on the creation of keyboard instruments designed to self-actuate. Trimpin's Vorsetzer (Focke, 2011) is an early example of such a mechatronic instrument; Godfried-Willem Raes uses a modified version of the Vorsetzer. Further, the Waseda Team developed an anthropomorphic self-playing robot. While not capable of such complicated augmentations as the aforementioned assistive playing devices, these automated augmentations allow composers a high degree of compositional flexibility over a piece's amplitude, melody, and harmonic content.

A widespread mechatronic instrument capable of a high degree of precision expressivity in the Yamaha Disklavier (and similar instruments), a device that can be viewed as a hybrid between assistive and automated keyboard players. The Disklaiver has a rich history of expressive and experimental compositions, with composers such as Horacio Vaggione (Budon, 2000), David Rosenboom, Terry Riley, Michael Pascal, and Morton Subotnick creating pieces for it. More recently, composer Jeff Aaron Bryant has explored the integration of new musical interface-driven performance with Disklaviers (Bryant, 2012).

New modes of human/instrument interaction in works such as those by Bryant highlight the creative affordances of modern mechatronic keyboard instruments. The new work presented below in this paper seeks to create a similar level of expressivity for a reed-based keyboard instrument.

2.3. Recent Advances in Robotic Percussion

As the most widely-built variety of musical robots, robotic percussion systems are used in most of the discipline's sub-fields. The abundance of robotic percussion instruments is likely due to their relative ease of implementation: potentially mechatronically-simple, robotic drummers can be built with few moving parts, allowing for workers to focus on compositional rather than engineering goals.

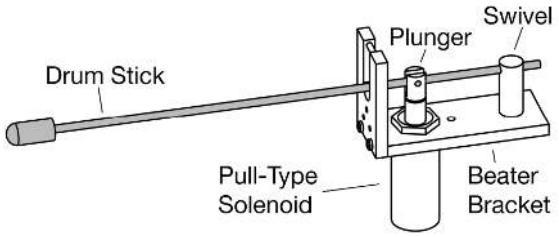


Figure 3. An example of a solenoid-based mechatronic drum beater.

(Murphy, 2012) and (Kapur, 2005) provide a detailed history of robotic percussion systems; this paper will focus briefly on those systems that attempt to further the expressive potential of mechatronic percussion devices.

Many existent robotic percussion systems use solenoid actuators with a single degree of freedom, configured to bring a drumstick or other effector into contact with a drumhead upon actuation of the solenoid. An exemplar device is illustrated in Figure 3. Such drumming systems are used in many contexts, including the authors' robotic ensembles (Kapur, 2011).

While existent solenoid-based drumming systems allow for easy implementation and simple driver electronics, they lack the ability to consistently vary their striking force (Murphy, June 2012) and

position above the drum head. Roboticist Gil Weinberg has built a system that addresses these shortcomings (Weinberg, 2006). Other approaches involve using many discrete solenoids (Kapur, 2011b) or (in the case of Trimpin and Godfried-Willem Raes) more complicated driver electronics.

Based upon the state of the art of mechatronic percussion, a need is perceived for an easy-to-build, inexpensive, compact, and easily-interfaced mechatronic drum system. This system is introduced in the following section.

3. NEW EXPRESSIVE MUSICAL ROBOTS

This section presents an overview of the current ensemble of expressive musical robots built by the authors through Victoria University of Wellington's Sonic Engineering program. While consisting of a diverse array of instruments, this ensemble was constructed with three common goals: to build robotic instruments that are more expressive than typical existent instruments, to provide composers with relatively means of interfacing with the robots, and to create them using inexpensive manufacturing techniques. This section presents five recently-completed instruments: two chordophones, a harmonium, and a mechatronic drum beater all designed to fulfil the above goals. The means by which the robots fulfil the goals are described, and brief overviews of the robots are provided.

3.1. Expressive Robotic Chordophones

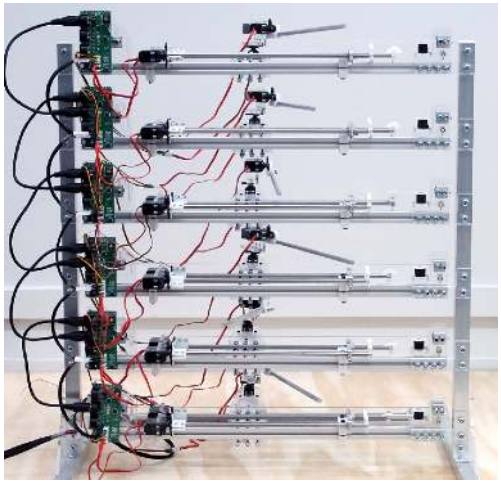


Figure 4. Swivel, a six-stringed robotic slide guitar.

In an effort to build expressive chordophones, a robotic slide guitar and bass guitar have been built. Both the slide guitar, dubbed Swivel, and the bass guitar, dubbed MechBass, are modular instruments: as shown in Figures 4 and 5, MechBass and Swivel consist of arrays of string modules. These modules are independently-addressable and allow each string to be composed for individually.

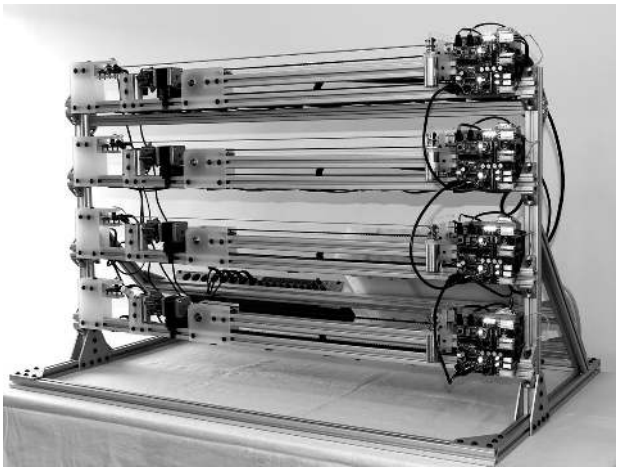


Figure 5. MechBass, a mechatronic bass guitar.

Both MechBass and Swivel 2 are assemblies of subsystems; each subsystem is designed to allow for a greater amount of compositional expressivity than is normal in most existent mechatronic chordophones.

To alter the pitch of a module's string, Swivel 2 and MechBass employ fretter mechanisms. After examining existing mechatronic chordophones, solenoid array fretting systems (described above in Section 2.1) were rejected in favour of motorised positioning systems. These positioning systems were chosen due to their ability to be used for vibrato, pitch bend, and alternative tuning schemes: such techniques extend the compositional expressivity of the instruments.

MechBass uses a fretter system similar to Eric Singer's GuitarBot (Singer, 2003), but with small changes implemented to increase the compositional flexibility and performance of the fretter. Where GuitarBot uses a linear motion slide permanently in contact with the string, the fretter system on MechBass can be engaged and disengaged from the string with the aid of solenoid-actuated clamps: by disengaging from the string, the fretter allows composers to choose whether a pitchshift event is a pitchbend or discrete step.

After the development of MechBass, it was noted that the use of a positioning device for pitch shift events introduced some degree of latency between fretting events: in the worst case, the fretter trolley must move the full length of the guitar's neck, introducing some delay into the fretting action. A fretter capable of very fast repositioning is therefore advantageous. It was a goal in the development of Swivel 2 to build such a fretter. Swivel 2's fretter, shown in Figure 6, is a rotary arm mounted on each module in such a manner that it can reach each position on the string at different motor shaft angles. Upon reaching the desired angle, the arm is pressed against the string by a second motor, changing the string's vibratory length.

The fretting mechanisms on both Swivel 2 and MechBass allow for further parametric control over fretting events than is possible with prior mechatronic chordophones. As such, they fulfil the initial goals of building mechatronic instruments endowed with a larger

number of expressive parameters than are present in previous mechatronic instruments.

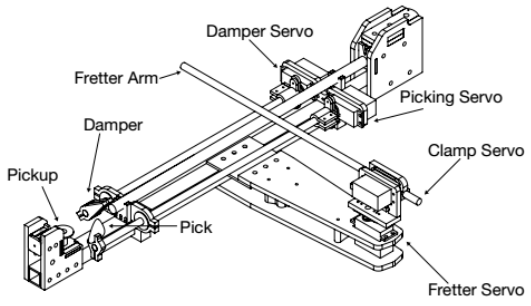


Figure 6. A drawing of a Swivel 2 module.

The second subsystem designed with an intention of enhanced expressivity for MechBass and Swivel 2 is the robots' picking mechanism. While Swivel 2 uses a reciprocating pickwheel similar to GuitarBot and other pickwheel-based systems, MechBass uses an enhanced pickwheel, designed to afford composers an additional parameter of control.

The pickwheel system on each module of MechBass can be raised and lowered in response to a composer's commands, changing the picks' proximity to the strings. By playing closer to the string, the pick's power increases, resulting in a louder, more powerful pick event. Softer, less powerful pick events can be achieved by lowering the pickwheel away from the string. To raise and lower the pickwheel, the stepper motor is mounted on a pivoting cam. The cam is connected to a miniature servomotor's shaft: as the servomotor rotates, the pickwheel, shown in Figure 7, pivots about its hinge.

Similarly, the damper mechanisms of both Swivel 2 and MechBass allow for more control than is typical of robotic chordophones. Where most dampers are on/off solenoids, MechBass and Swivel use rotary servomotors, able to vary the amount of force with which they press the string.

To retain compatibility with the authors' other musical robots, Swivel 2 and MechBass respond to MIDI messages from a host device. Each of the MechBass and Swivel 2 modules is assigned a distinct MIDI channel, and each actuator subassembly responds to distinct MIDI commands. The commands' parameters are converted by the modules' microcontrollers to output voltages to interface with PCB assembly-mounted motor drivers.

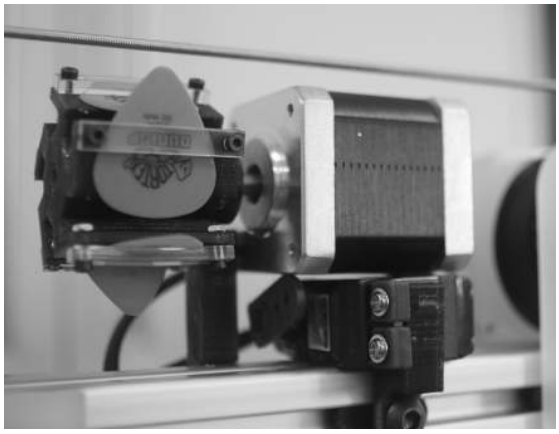


Figure 7. The adjustable-height pickwheel used on each MechBass module.

3.2. An Expressive Robotic Harmonium



Figure 8. Kritaanjli, a mechatronically-augmented harmonium.

In building a new mechatronic keyboard-playing instrument, a portable desktop instrument endowed with the most expressive parameters of the existent surveyed mechatronic keyboard instruments was desired. As discussed in Section 2.2, contemporary augmented pianos are capable of tremolo (volume modulation) and extended polyphony. These two parameters were chosen to be added to a harmonium, an instrument used abundantly in Indian folk and devotional music (a genre explored by the second author).

Like MechBass and Swivel 2, Kritaanjli is best viewed as an assemblage of mechatronic systems, positioned around a musical instrument to allow for computer-controlled actuation events. Kritaanjli is made up of two such assemblies: a keyboard playing mechanism and a bellows-pumping system. The remainder of this subsection discusses the means by which both subsections are constructed to afford compositional expressivity.

To allow for the realisation of harmonically-complicated compositions, a parallel approach is used by Kritaanjli's keyboard playing system: above each key is

a linear solenoid actuator, switched by a power MOSFET. A sufficiently large DC power supply is used to allow for all 44 keys to be depressed simultaneously. Further, the solenoids may be cycled quite quickly, allowing for the playback of new extended technique compositions.

Harmoniums produce sound whose loudness corresponds to the rate at which the instrument's bellows are pumped. Kritaanjli is capable of variable-loudness playing by varying the speed with which its bellows are automatically pumped. The automatic pumping mechanism consists of a DC gearmotor driving a crankshaft. The shaft's reciprocal motion is connected to a hinge on the instrument's bellows, allowing it to be moved back and forth in response to the crankshaft's displacement. To achieve expressive volume changes, the motor's speed can be varied.

Like Swivel 2 and MechBass, Kritaanjli responds to MIDI commands sent from a host computer. The keyboard-playing solenoids are switched on in response to NoteOn events and disabled upon the arrival of NoteOff commands. The bellows-pumping speed is adjusted by varying the value of a MIDI Control Change (CC) 7 message.

The combination of full-keyboard polyphony and adjustable output dynamics allows for Kritaanjli to be used in a compositionally-expressive manner: by providing composers with a relatively large number of adjustable parameters, Kritaanjli, like the other mechatronic instruments described in this paper, allows composers increased freedom over their compositional decisions.

3.3. A Parametrically-Rich Mechatronic Drum Player

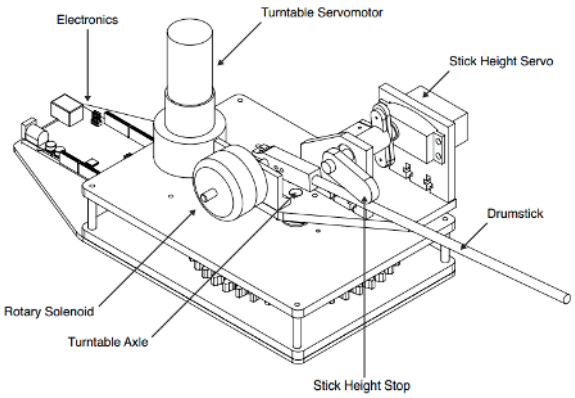


Figure 9. A drawing of Nudge, illustrating its rotary drumstick-beating mechanism and adjust height stop.

In creating an ensemble of musical robotics designed to allow for increased compositional expressivity, it was deemed important to include in the ensemble a mechanism capable of striking drums or other percussive objects. The resulting mechatronic instrument, Nudge (a drawing of which is shown in Figure 9), is "parametrically-rich," allowing composers more control over the percussion system's parameters than is normal for existent mechatronic drum beaters.

Like the majority of existent drum beaters, Nudge uses a solenoid to rapidly bring a drumstick into contact with a drum or other percussive object. Other actuator types were rejected due to low speed or high actuation noise. While most pre-existing mechatronic drum beaters are limited to the single solenoid actuator pre-positioned above a drum head, Nudge provides two more parameters to allow for increased compositional expressivity: Nudge's drumstick may be rotated above one or more percussive objects and its drumstick's at-rest height can be adjusted on the fly.

By being able to rotate its attached drumstick above the drum head, Nudge allows composers to select which part of an object will be hit: for example, Nudge can engage in rimshots at the start of a bar, before repositioning and hitting the centre of the drum at the end of the bar. While (Weinberg, 2006) allows for repositioning of the drumstick, Nudge accomplishes such positioning with a greatly simplified mechanical system, requiring no linear motion hardware.

A recurring problem with many mechatronic drum systems (particularly those based upon solenoid actuators) is that of drumstick positioning in relation to the drum head. If the drumstick's at-rest position is too far from the drum head, rapid, soft percussive events will not move the drumstick's solenoid sufficiently to bring it into contact with the drum. Conversely, if mounted too close to the drumhead, the drumstick will be unable to gain sufficient inertia to allow for powerful, loud strike events. To address this problem, Nudge's drumstick's at-rest position can be varied by a servo-attached cam mechanism. As the cam rotates, the servo pushes the at-rest drumstick from a position high above the drum head (suitable for slower, louder strikes) to one quite close to the drum head (suitable for rapid rolls and quiet strike events).

Like Kritaanjli, Swivel, and MechBass, Nudge communicates with a host device with MIDI. NoteOn and NoteOff commands enable and disable the drum-beater solenoid. A closed-loop control scheme is used to rotate the turntable: a MIDI pitch bend command serves as a setpoint, and an internal PI controller uses readings from the turntable's DC servomotor to reach the setpoint. Drumstick stop height is adjusted in response to MIDI CC 7 commands.

As with the other robots described in this paper, Nudge allows composers to expressively realise compositional ideas that would have been difficult with traditional mechatronic drum beaters (such as those described in (Kapur, 2005)). By allowing the drum-striking solenoid to be flexibly positioned above the drum head, a single Nudge drum player takes the place of a large number of static drum beaters.



Figure 10. Nudge, positioned to strike a frame drum.

4. INTERFACING WITH EXPRESSIVE MUSICAL ROBOTS

The large number of user-controllable parameters made available on MechBass, Swivel 2, Kritaanjli, and Nudge present a potential problem: as the number of parameters increases, the complexity of interfacing with an instrument increases (a topic further explored in (Dobrian, 2006)). To facilitate not only the robots' parametric density but also their expressive potential, Tangle, a robot networking software suite, was designed and built with the aid of VUW student Paul Mathews.

Tangle, implemented in the ChucK programming language (Wang, 2008), allows for a client computer to connect to a server to which are attached any number of musical robotic devices. While the robots may have a plethora of interfacing options, the server maps them to a device-agnostic interface, presenting the client with an array of similar-looking instruments: where two robots might require different low-level messages, Tangle allows Clients to interact with them at a higher level, allowing composers unfamiliar with the robots to more rapidly begin exploring their expressive parameters.

Further, Tangle allows instrument-specific "macros" to be programmed, allowing for a single message from a client computer to instigate a potentially complicated sequence of actions on the server.

While a detailed discussion of Tangle's architecture lies outside the scope of this overview, the following paragraphs detail three exemplar features of Tangle, deemed representative of the open-ended nature of software-based augmentation able to be performed on Tangle-connected robots.

Further information about Tangle, including detailed descriptive documentation and source code, may be found at https://github.com/PFCM/Tangle.

5. CONCLUSIONS

A fast-growing field, mechatronic and robotic music systems are in need of increased expressivity. This expressivity is best achieved through two means: enhanced parametric control of the mechatronic systems and improved means of interfacing with them.

The work presented in this paper endeavours to improve both: new systems with increased degrees of actuator freedom are presented, alongside a software suite intended to allow for increased ease of interfacing with them.

Current work on the mechatronic systems focuses on valuating composers' experiences with them: user studies are in progress wherein composers are asked to write pieces for the robots that explore their expressive parameters; the composers are asked to evaluate their experiences with the new systems.

While significant work still lies ahead in both the advancement of parametric complexity of the robots and enhanced ease-of-interface between the robots and users, the work presented in this paper advances the state of the art, pointing toward a future of expressive and intuitive mechatronic instruments.

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